

CARP Meeting #3

Implement M

How it works

MUL_INIT

INPUTS:

- 32 bit RS1
- 32 bit RS2
- 2 bit OPCODE

- ◆ MUL = 00
- ◆ MULH = 01
- ◆ MULSHU = 10
- ◆ MULHU = 11

OUTPUTS:

- 64 bit output PROD
- Done flag
- signed

MUL_MID

INPUTS:

- signed
- mul_done
- 64 bit INTERMEDIATE
- 2 bit OPCODE

- ◆ MUL
- ◆ MULH
- ◆ MULSHU
- ◆ MULHU

OUTPUTS:

- 64 bit output PROD
- Done flag
- Signed

MUL_FIN

INPUTS:

- signed
- mul_done
- 64 bit INTERMEDIATE
- 2 bit OPCODE

- ◆ MUL
- ◆ MULH
- ◆ MULSHU
- ◆ MULHU

- 32 bit output RESULT

